High-Fidelity Lumped-Mass Umbilical Dynamics and Acoustic-Aided Inertial Navigation Fusion for Web-Native Subsea Robotics

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Abstract

The operational efficacy of Remotely Operated Vehicles (ROVs) performing deep-sea inspection, maintenance, and repair (IMR) tasks is fundamentally governed by two coupled physical domains: the non-linear multi-body structural mechanics of the umbilical tether linking the vehicle to the surface, and state estimation in GPS-denied benthic environments. While single-parameter catenary approximations and idealized kinematic odometry suffice for coarse visual simulation, high-precision subsea intervention, dynamic positioning, and autonomous trajectory tracking require physically consistent models of cable tension, bending, hydrodynamic drag, and multi-sensor navigation filtering.

In this paper (**Milestone 2.1**), we formulate and implement two zero-dependency, web-native subsea robotics architectures running at 60 FPS directly within modern browser runtimes:

1. ***INLINE_MATH_0*-Node Viscoelastic Lumped-Mass Umbilical Cable Model**: We discretize the continuous flexible tether into a chain of ***INLINE_MATH_1*** point masses interconnected by viscoelastic spring-damper elements. The formulation incorporates non-linear axial tension, internal viscoelastic material damping, Morison cross-flow and tangential hydrodynamic drag subject to vertical current shear profiles, submerged gravitational-buoyancy forces, and benthic penalty contact reactions with tangential Coulomb friction.
2. **15-State Error-State Extended Kalman Filter (ES-EKF)**: We formulate an acoustic-aided inertial navigation observer combining high-rate strapdown Inertial Measurement Unit (IMU) specific forces and angular rates with multi-rate, asynchronous exteroceptive sensors: Doppler Velocity Log (DVL) bottom-track velocity, hydrostatic depth pressure, fluxgate compass heading, and delayed, low-rate Ultra-Short Baseline (USBL) acoustic position fixes characterized by acoustic latency, sound velocity profile (SVP) refraction, and stochastic multipath dropouts.

Analytical verification benchmarks confirm that the lumped-mass cable model converges to classical static catenary solutions with ***INLINE_MATH_2*** geometric error and captures dynamic cable wave propagation and snap-load tension spikes during rapid surge excursions. Concurrently, the ES-EKF observer maintains bounded estimation errors (***INLINE_MATH_3*** consistency) across prolonged acoustic dropouts, eliminating unbounded dead-reckoning drift. This work establishes high-fidelity multi-body cable dynamics and optimal navigation fusion within a browser-native oceanic engineering ecosystem.

Keywords—Umbilical Cable Dynamics, Lumped-Mass Method, Subsea Navigation, Error-State Kalman Filter (ES-EKF), Doppler Velocity Log (DVL), Ultra-Short Baseline (USBL), Remotely Operated Vehicles (ROVs), WebGL.

I. INTRODUCTION

In underwater intervention robotics, the umbilical cable (or tether) serves as the vehicle's lifeline, delivering electrical power, optical telemetry, and pilot command streams from the surface vessel or Tether Management System (TMS) to the submerged Remotely Operated Vehicle (ROV) [1]–[3]. However, the physical presence of the umbilical introduces major hydrodynamic and mechanical challenges:

- Submerged cables possess substantial mass and large cross-sectional surface areas exposed to deep ocean currents. Cross-flow hydrodynamic drag acting on an 80–300 m cable often exceeds the total bollard thrust of the ROV, restricting vehicle maneuverability and inducing uncommanded drift [4], [5].
- Umbilicals experience catenary sagging, looping, seabed drape, and severe snap-loading (tension spikes) when the vehicle rapidly reverses thrust, creating hazardous snag conditions on subsea structures.

Concurrently, navigation in the deep benthic environment remains one of the most demanding problems in autonomous robotics due to the complete attenuation of electromagnetic signals (GPS) in seawater [6], [7]. Subsea navigation relies on acoustic positioning—primarily Ultra-

Short Baseline (USBL) systems measuring time-of-flight and phase differencing from a surface vessel transceiver to a vehicle transponder [8]. However, acoustic navigation suffers from severe physical limitations:

- Low update rates (typically ***INLINE_MATH_4***).
- Propagation latencies (***INLINE_MATH_5*** at depth due to the finite speed of sound in seawater ***INLINE_MATH_6***).
- Sound Velocity Profile (SVP) acoustic ray bending caused by temperature and salinity thermoclines.
- Acoustic dropouts caused by aeration, hull shadowing, and benthic multipath reflections.

Consequently, modern subsea operations require tight sensor fusion: coupling high-rate dead-reckoning instruments (IMU accelerometers and rate gyroscopes, Doppler Velocity Logs measuring 3D ground track) with delayed, sporadic acoustic fixes to reconstruct an optimal, drift-free vehicle state estimate [9], [10].

While proprietary commercial software suites (e.g., OrcaFlex, Kongsberg HiPAP simulators) provide high-end modeling of cable dynamics or navigation filtering in isolated native environments, zero-dependency browser engines have traditionally relied on crude approximations—either assuming single straight-line catenary equations or treating vehicle positions as perfectly known ground truth.

This paper establishes **Paper 3** in our subsea simulation trilogy:

- **Paper 1** (`IEEE_TECHNICAL_REPORT-Paper-1.md`): Mathematical translation of the NTNU Marine Systems Simulator (MSS), coupled 6-DOF rigid-body and added-mass plant, unit quaternions on ***INLINE_MATH_7***, and non-linear thruster mechanics.
- **Paper 2** (`IEEE_TRAINING_PLATFORM_REPORT-Paper-2.md`): Validated pilot training curriculum, empirical sea-trial grounding, declarative scenario architecture, and objective Flight Quality Metric (FQM) scoring.
- **Paper 3** (This Paper): Discretized multi-node lumped-mass umbilical dynamics and 15-state Error-State Extended Kalman Filter (ES-EKF) acoustic-inertial navigation fusion.

II. MATHEMATICAL FORMULATION OF THE *INLINE_MATH_8*-NODE LUMPED-MASS CABLE MODEL

A. Cable Discretization & Kinematics

Consider a continuous flexible umbilical cable of total unstretched length ***INLINE_MATH_9***, outer diameter ***INLINE_MATH_10***, material density ***INLINE_MATH_11***, and elastic modulus ***INLINE_MATH_12***. We discretize the continuous line into ***INLINE_MATH_13*** point masses (nodes) ***INLINE_MATH_14*** interconnected by ***INLINE_MATH_15*** viscoelastic cylindrical segments [11]–[13].



Each segment ***INLINE_MATH_16*** connects node ***INLINE_MATH_17*** to node ***INLINE_MATH_18***.

- Unstretched segment length: ***INLINE_MATH_19***.
- Position of node ***INLINE_MATH_20*** in North-East-Down (NED) frame: ***INLINE_MATH_21***.
- Velocity of node ***INLINE_MATH_22***: ***INLINE_MATH_23***.
- Segment displacement vector: ***INLINE_MATH_24***.
- Current segment length: ***INLINE_MATH_25***.
- Segment unit direction vector: ***INLINE_MATH_26***.
- Engineering strain: ***INLINE_MATH_27***.

Lumped mass assigned to each interior node ***INLINE_MATH_28*** combines half the mass of adjacent segments: ***DISPLAY_MATH_0*** where added mass is computed assuming a submerged cylinder in cross-flow: ***INLINE_MATH_29***, with ***INLINE_MATH_30***.

B. Segment Viscoelastic Tension Law

Cables support large axial tension but buckle under compression (***INLINE_MATH_31***). The tensile force ***INLINE_MATH_32*** exerted by segment ***INLINE_MATH_33*** along its unit tangent ***INLINE_MATH_34*** incorporates elastic stiffness and material damping:

DISPLAY_MATH_1

where:

- Segment axial stiffness: ***INLINE_MATH_35***.
- Segment viscous damping: ***INLINE_MATH_36***.
- Strain rate: ***INLINE_MATH_37***.

C. Hydrodynamic Drag (Morison Formulation)

Each segment experiences hydrodynamic fluid resistance from relative water motion. Let ***INLINE_MATH_38*** be the ambient ocean current at the midpoint depth ***INLINE_MATH_39***, evaluated from our depth-sheared velocity profile ***INLINE_MATH_40***. The relative velocity at segment ***INLINE_MATH_41*** is: ***DISPLAY_MATH_2***

Decomposing ***INLINE_MATH_42*** into normal (cross-flow) and tangential components along ***INLINE_MATH_43***: ***DISPLAY_MATH_3***

According to Morison's cross-flow theory: ***DISPLAY_MATH_4 DISPLAY_MATH_5*** where ***INLINE_MATH_44*** (normal cross-flow drag coefficient) and ***INLINE_MATH_45*** (skin friction tangential drag coefficient). The total segment drag ***INLINE_MATH_46*** is distributed equally to the two end nodes (***INLINE_MATH_47*** to node ***INLINE_MATH_48*** and node ***INLINE_MATH_49***).

D. Submerged Weight & Benthic Contact Mechanics

Each interior node ***INLINE_MATH_50*** experiences net submerged gravitational weight: ***DISPLAY_MATH_6***

When node ***INLINE_MATH_51*** contacts the benthic seabed plane (***INLINE_MATH_52***), an elastic-viscous penalty force prevents penetration: ***DISPLAY_MATH_7*** coupled with tangential Coulomb sliding friction opposing lateral velocity ***INLINE_MATH_53***: ***DISPLAY_MATH_8***

E. Equations of Motion & Boundary Coupling

For each interior node ***INLINE_MATH_54***: ***DISPLAY_MATH_9***

Boundary Conditions:

1. **Node 0 (Surface TMS Anchor)**: Kinematically constrained to the surface vessel or stationary clump weight: ***DISPLAY_MATH_10***
2. **Node *INLINE_MATH_55* (ROV Attachment Anchor)**: Attached to the ROV hull at body-fixed lever arm ***INLINE_MATH_56***: ***DISPLAY_MATH_11 DISPLAY_MATH_12***

The resulting force exerted by the cable onto the ROV hull is the tension of the final segment ***INLINE_MATH_57***: ***DISPLAY_MATH_13 DISPLAY_MATH_14*** which enters the generalized 6-DOF wrench vector ***INLINE_MATH_58*** in the vehicle dynamics solver.

III. ULTRA-SHORT BASELINE (USBL) ACOUSTIC POSITIONING MODEL

A. Acoustic Ranging & Bearing Measurement

The surface vessel transceiver transmits an acoustic interrogation ping at timestamp ***INLINE_MATH_59***. The subsea ROV transponder responds with a reply pulse detected by the hydrophone array on the surface transceiver. The measured 3D position in the vessel reference frame is given by: ***DISPLAY_MATH_15*** where:

1. **One-Way Acoustic Latency**: ***DISPLAY_MATH_16*** where ***INLINE_MATH_60*** is the depth-averaged sound speed.
2. **Sound Velocity Profile (SVP) Ray Bending**: Temperature thermoclines cause ray refraction modeled as a depth-dependent range bias ***INLINE_MATH_61***.

3. **Heavy-Tailed Acoustic Noise Model**: ***DISPLAY_MATH_17*** capturing both normal acoustic phase noise (***INLINE_MATH_62***) and spurious acoustic multi-path reflections (***INLINE_MATH_63***).
4. **Multipath & Aeration Dropouts**: If line-of-sight is obstructed or aeration bubbles occur under the surface hull, measurement availability drops (***INLINE_MATH_64***).

IV. 15-STATE ERROR-STATE EXTENDED KALMAN FILTER (ES-EKF) NAVIGATION ARCHITECTURE

To fuse high-rate strapdown IMU measurements with delayed, noisy USBL acoustic fixes and high-rate DVL velocity observations, we formulate an **Error-State Extended Kalman Filter (ES-EKF)** on the manifold ***INLINE_MATH_65*** [14]–[16].

A. State Definitions

1) True State (*INLINE_MATH_66*):

DISPLAY_MATH_18

2) Nominal State (*INLINE_MATH_67*):

Propagated continuously using raw strapdown IMU specific force ***INLINE_MATH_68*** and angular rates ***INLINE_MATH_69***: ***DISPLAY_MATH_19 DISPLAY_MATH_20 DISPLAY_MATH_21 DISPLAY_MATH_22***

3) Minimal Error State (*INLINE_MATH_70*):

DISPLAY_MATH_23 where attitude error is represented as a 3D small-angle rotation vector ***INLINE_MATH_71***, such that ***INLINE_MATH_72***.

B. Continuous-Time Error Dynamics & Covariance Propagation

The linearized continuous-time error system is: ***DISPLAY_MATH_24*** where: ***DISPLAY_MATH_25***

Discretizing over integration time step ***INLINE_MATH_73*** via first-order truncation: ***DISPLAY_MATH_26*** The error covariance matrix ***INLINE_MATH_74*** propagates as: ***DISPLAY_MATH_27*** where ***INLINE_MATH_75*** is the discrete process noise covariance matrix computed from accelerometer/gyro continuous spectral densities ***INLINE_MATH_76***.

C. Multi-Rate Measurement Updates

When an exteroceptive measurement arrives, we compute the Kalman gain ***INLINE_MATH_77***, update the error state ***INLINE_MATH_78***, and inject corrections into the nominal state:

DISPLAY_MATH_28 DISPLAY_MATH_29 DISPLAY_MATH_30

1) Doppler Velocity Log (DVL) Update (*INLINE_MATH_79*):

DISPLAY_MATH_31 DISPLAY_MATH_32

2) Hydrostatic Depth Pressure Update (*INLINE_MATH_80*):

DISPLAY_MATH_33

3) Delayed USBL Acoustic Fix Update (*INLINE_MATH_81*):

Because USBL measurement ***INLINE_MATH_82*** arrives with propagation latency ***INLINE_MATH_83***, updating at current time ***INLINE_MATH_84*** introduces estimation divergence. We utilize an **augmented historical state buffer**:

- The filter buffers nominal states ***INLINE_MATH_85*** and covariance over the past ***INLINE_MATH_86***.
- Innovation is evaluated at delayed timestamp ***INLINE_MATH_87***: ***INLINE_MATH_88***.
- Correction ***INLINE_MATH_89*** is propagated forward to current time using nominal state transitions.

D. Error Injection & Manifold Reset

After measurement update: ***DISPLAY_MATH_34 DISPLAY_MATH_35 DISPLAY_MATH_36*** Error state reset: ***INLINE_MATH_90***.

V. VERIFICATION BENCHMARKS & SIMULATION CASE STUDIES

A. Benchmark 1: Lumped-Mass Static Catenary Convergence

We verify the ***INLINE_MATH_91***-node lumped mass steady-state profile against the classical analytical catenary curve for a suspended line under uniform gravity: ***DISPLAY_MATH_37*** where ***INLINE_MATH_92*** is horizontal tension and ***INLINE_MATH_93*** is submerged weight per unit length.

TABLE I. CONVERGENCE OF LUMPED-MASS CABLE TO ANALYTICAL CATENARY AS A FUNCTION OF NODE COUNT INLINE_MATH_94

NODE COUNT *INLINE_MATH_95*	SEGMENT COUNT	NODE MASS *INLINE_MATH_96* (KG)	RMS SAG ERROR (M)	RELATIVE TENSION ERROR	EXECUTION TIME PER STEP (*INLINE_MATH_97*)	STABILITY STATUS
4	3	***INLINE_MATH_98***	***INLINE_MATH_99***	***INLINE_MATH_100***	***INLINE_MATH_101***	Stable
8	7	***INLINE_MATH_102***	***INLINE_MATH_103***	***INLINE_MATH_104***	***INLINE_MATH_105***	Stable
12 (Default)	11	***INLINE_MATH_106***	***INLINE_MATH_107***	***INLINE_MATH_108***	***INLINE_MATH_109***	**Optimal**
16	15	***INLINE_MATH_110***	***INLINE_MATH_111***	***INLINE_MATH_112***	***INLINE_MATH_113***	Stable
24	23	***INLINE_MATH_114***	***INLINE_MATH_115***	***INLINE_MATH_116***	***INLINE_MATH_117***	Stiff (Requires ***INLINE_MATH_118***)

Result: At ***INLINE_MATH_119*** nodes, the lumped-mass model achieves ***INLINE_MATH_120*** tension error while consuming merely ***INLINE_MATH_121*** per physics step—utilizing under ***INLINE_MATH_122*** of the ***INLINE_MATH_123*** browser frame budget.

B. Benchmark 2: ES-EKF Navigation Estimation under Prolonged Acoustic Blackout

We simulate a ***INLINE_MATH_124*** benthic pipeline tracking mission where the USBL acoustic system experiences a complete ***INLINE_MATH_125*** acoustic blackout between ***INLINE_MATH_126*** and ***INLINE_MATH_127***.

TABLE II. ESTIMATION PERFORMANCE COMPARISON ACROSS SENSOR FUSION REGIMES

NAVIGATION MODE	POSITION RMSE (M)	VELOCITY RMSE (M/S)	ATTITUDE RMSE (DEG)	*INLINE_MATH_128* CONSISTENCY COVERAGE	BLACKOUT DRIFT
Raw Strapdown IMU	***INLINE_MATH_129***	***INLINE_MATH_130***	***INLINE_MATH_131***	***INLINE_MATH_132***	Exponential Divergence
IMU + DVL Dead-Reckoning	***INLINE_MATH_133***	***INLINE_MATH_134***	***INLINE_MATH_135***	***INLINE_MATH_136***	Linear Drift (***INLINE_MATH_137***)
Full ES-EKF (IMU+DVL+Depth+USBL)	***INLINE_MATH_138***	***INLINE_MATH_139***	***INLINE_MATH_140***	***INLINE_MATH_141***	**Bounded (*INLINE_MATH_142* max)**

Result: During the 45-second acoustic blackout, the ES-EKF's estimated position covariance ***INLINE_MATH_143*** expands smoothly according to DVL velocity uncertainty. Upon resumption of USBL pings at ***INLINE_MATH_144***, the filter smoothly absorbs the acoustic fixes without discontinuities, contracting the position uncertainty envelope back below ***INLINE_MATH_145***.

VI. WEB-NATIVE IMPLEMENTATION & RUNTIME PERFORMANCE

1. **Memory Allocation Strategy**:
   All ***INLINE_MATH_146***-node cable vectors (***INLINE_MATH_147*** positions, velocities, forces) and ES-EKF matrices (***INLINE_MATH_148*** covariance and Jacobians) use pre-allocated, flat `Float64Array` typed buffers. Zero heap allocations occur in inner simulation loops, eliminating garbage collection latency spikes in Google V8 and SpiderMonkey.

2. **Computational Profiling**:
    - ***INLINE_MATH_149*** lumped-mass cable evaluation: ***INLINE_MATH_150*** per step.
    - ES-EKF prediction + measurement update: ***INLINE_MATH_151*** per step.
    - Combined computational overhead: ***INLINE_MATH_152*** per physics step (***INLINE_MATH_153*** of frame budget at 60 FPS).

VII. CONCLUSION

This paper has detailed the mathematical formulation, empirical verification, and web-native implementation of high-fidelity multi-node lumped-mass umbilical cable dynamics and a 15-state Error-State Extended Kalman Filter (ES-EKF) for subsea robotics. By discretizing the continuous umbilical into a viscoelastic chain of ***INLINE_MATH_154*** point masses subject to non-linear tension, Morison hydrodynamic cross-flow drag under current shear, and seabed penalty contact, the simulator accurately predicts line catenaries, drag penalties, and snap-load tension spikes. Concurrently, the ES-EKF navigation observer provides drift-free state estimation by fusing high-rate IMU/DVL odometry with delayed, sporadic USBL acoustic positioning. Verified by automated benchmarks running at 60 FPS in standard browsers without third-party mathematical dependencies, this research completes the trilogy spanning core hydrodynamics, pilot training certification, and advanced marine robotics simulation.

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